Code No: 158DE **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, July/August - 2022
ENERGY SOURCES AND APPLICATIONS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1. Explain the main sources of energy for power generation and discuss their sustainability. [15]

2. Explain the scientific principles and historical interpretation of energy use with respect to societal, environmental and climate issues. [15]

3.a) Explain the transformations of various energy sources into power generation.
b) Describe the processes of coal gasification and explain the outputs of the process. [7+8]

4. Discuss the remedies and alternatives for fossil fuels and explain their advantages and limitations. [15]

5. Explain the energy regeneration using Pumped storage system hydro power projects. [15]

6. Explain the process of superconductor-based energy storage system along with its merits and demerits. [15]

7. Discuss the economics of energy with respect to
a) Production and consumption
b) Environmental outcomes. [8+7]

8. Explain the concept of Green building and Green architecture along with their applications. [15]

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